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WDV442-O Module 3: Database Abstraction Layering

Assignment: Space Object Library API

Repository Link: <https://github.com/OlivaresStephanie-FS/ASL/tree/assignments/week-3-space-tracker>

Demonstrating CRUD operations for Galaxies, Stars, and Planets using cURL, Express, MySQL, Docker, and Sequelize ORM. It also demonstrates the required one-to-many and many-to-many Sequelize associations.

Requirements Demonstrated:

- Galaxy resource: Create, Read, Update, Delete
- Star resource: Create, Read, Update, Delete
- Planet resource: Create, Read, Update, Delete
- Galaxy has many Stars
- Star belongs to Galaxy
- Star belongs to many Planets
- Planet belongs to many Stars
- StarsPlanets join table used for the Star/Planet many-to-many relationship

API Baseline and Empty Resource Checks:

This screenshot shows the API responding successfully at localhost:3000 and the initial resource endpoints returning empty arrays before new records are created.

```
$ curl http://localhost:3000/galaxies
curl http://localhost:3000/stars
curl http://localhost:3000/planets[]$ []$
[]$ curl -i http://localhost:3000/
HTTP/1.1 200 OK
X-Powered-By: Express
Content-Type: text/html; charset=utf-8
Content-Length: 31
ETag: W/"1f-niBwj1bbA50eyQ5XyDLszDIm/CI"
Date: Sat, 20 Jun 2026 02:11:49 GMT
Connection: keep-alive
Keep-Alive: timeout=5

Welcome to Star Tracker Library$ curl -i http://localhost:3000/galaxies
HTTP/1.1 200 OK
X-Powered-By: Express
Content-Type: application/json; charset=utf-8
Content-Length: 2
ETag: W/"2-l9Fw4VU07kr8CvBl4zaMCqXZ0w"
Date: Sat, 20 Jun 2026 02:12:00 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

CREATE Operations: Galaxy and Star

This screenshot demonstrates POST requests creating the Milky Way galaxy and the Sun star. The Star record is linked to the Galaxy using GalaxyId.

```
[ ]$ curl -X POST -H "Content-Type: application/json" --data '{"name":"Milky Way","size":100000,"description":"The galaxy that contains our solar system."}' http://localhost:3000/galaxies
{"id":1,"name":"Milky Way","size":100000,"description":"The galaxy that contains our solar system.","updatedAt":"2026-06-20T02:12:42.822Z","createdAt":"2026-06-20T02:12:42.822Z"}$ curl http://localhost:3000/galaxies
[{"id":1,"name":"Milky Way","size":100000,"description":"The galaxy that contains our solar system.","createdAt":"2026-06-20T02:12:42.000Z","updatedAt":"2026-06-20T02:12:42.000Z"}]$ curl -X POST -H "Content-Type: application/json" --data '{"name":"Sun","size":1392000,"description":"The star at the center of our solar system.","GalaxyId":1}' http://localhost:3000/stars
```

CREATE Operations: Planet and StarsPlanets Join Record

This screenshot demonstrates creating the Earth planet and then creating the StarsPlanets join-table record linking StarId 1 to PlanetId 1.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL ... docker + - [ ] [ ] ... | X
→ assignment git:(main) x docker exec -it assignment-wdv442-node-1 sh

$ curl -X POST \
-H "Content-Type: application/json" \
--data '{"name":"Earth","size":12742,"description":"Third planet from the Sun."}' \
http://localhost:3000/planets> >
{"id":1,"name":"Earth","size":12742,"description":"Third planet from the Sun.","updatedAt":"2026-06-20T02:18:13.589Z","createdAt":"2026-06-20T02:18:13.589Z"}$ curl -s http://localhost:3000/planets | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
[
  {
    "id": 1,
    "name": "Earth",
    "size": 12742,
    "description": "Third planet from the Sun.",
    "createdAt": "2026-06-20T02:18:13.000Z",
    "updatedAt": "2026-06-20T02:18:13.000Z",
    "Stars": []
  }
]
$ node -e "const { StarsPlanets } = require('./models'); StarsPlanets.create({ StarId: 1, PlanetId: 1 }).then(console.log).catch(console.error)"
Executing (default): INSERT INTO `StarsPlanets` (`StarId`,`PlanetId`,`createdAt`,`updatedAt`) VALUES (?, ?, ?, ?);
StarsPlanets {
  dataValues: {
    StarId: 1,
    PlanetId: 1,
    updatedAt: 2026-06-20T02:24:37.117Z,
    createdAt: 2026-06-20T02:24:37.117Z
  },
  _previousDataValues: {
    StarId: 1,
    PlanetId: 1,
    createdAt: 2026-06-20T02:24:37.117Z,
    updatedAt: 2026-06-20T02:24:37.117Z
  },
  uniqno: 1,
  _changed: Set(0) {},
  _options: {
    isNewRecord: true,
    _schema: null,
    _schemaDelimiter: '',
    attributes: undefined,
    include: undefined,
    raw: undefined,
    silent: undefined
  },
  isNewRecord: false
}
```

READ Operation: GET /galaxies

This screenshot demonstrates reading all galaxies. The response includes the Milky Way galaxy and its associated Sun star.

```
$ curl -s http://localhost:3000/galaxies | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
[
  {
    "id": 1,
    "name": "Milky Way",
    "size": 100000,
    "description": "The galaxy that contains our solar system.",
    "createdAt": "2026-06-20T02:12:42.000Z",
    "updatedAt": "2026-06-20T02:12:42.000Z",
    "Stars": [
      {
        "id": 1,
        "name": "Sun",
        "size": 1392000,
        "description": "The star at the center of our solar system.",
        "GalaxyId": 1,
        "createdAt": "2026-06-20T02:13:33.000Z",
        "updatedAt": "2026-06-20T02:13:33.000Z"
      }
    ]
  }
]
```

READ Operation: GET /stars

This screenshot demonstrates reading all stars. The response includes the Sun star, its associated Galaxy, and the Planets array.

```
$ curl -s http://localhost:3000/stars | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
[
  {
    "id": 1,
    "name": "Sun",
    "size": 1392000,
    "description": "The star at the center of our solar system.",
    "GalaxyId": 1,
    "createdAt": "2026-06-20T02:13:33.000Z",
    "updatedAt": "2026-06-20T02:13:33.000Z",
    "Galaxy": {
      "id": 1,
      "name": "Milky Way",
      "size": 100000,
      "description": "The galaxy that contains our solar system.",
      "createdAt": "2026-06-20T02:12:42.000Z",
      "updatedAt": "2026-06-20T02:12:42.000Z"
    },
    "Planets": []
  }
]
```

READ Operation: GET /planets

This screenshot demonstrates reading all planets. The response includes Earth and its associated Star through the StarsPlanets relationship.

```
$ curl -s http://localhost:3000/planets | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
[
  {
    "id": 1,
    "name": "Earth",
    "size": 12742,
    "description": "Third planet from the Sun.",
    "createdAt": "2026-06-20T02:18:13.000Z",
    "updatedAt": "2026-06-20T02:18:13.000Z",
    "Stars": [
      {
        "id": 1,
        "name": "Sun",
        "size": 1392000,
        "description": "The star at the center of our solar system.",
        "GalaxyId": 1,
        "createdAt": "2026-06-20T02:13:33.000Z",
        "updatedAt": "2026-06-20T02:13:33.000Z",
        "StarsPlanets": {
          "StarId": 1,
          "PlanetId": 1,
          "createdAt": "2026-06-20T02:24:37.000Z",
          "updatedAt": "2026-06-20T02:24:37.000Z"
        }
      }
    ]
  }
]
```

READ ONE Operation: GET /galaxies/1

This screenshot demonstrates reading one Galaxy by ID and shows the associated Stars array.

```
$ curl -s http://localhost:3000/galaxies/1 | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
{
  "id": 1,
  "name": "Milky Way",
  "size": 100000,
  "description": "The galaxy that contains our solar system.",
  "createdAt": "2026-06-20T02:12:42.000Z",
  "updatedAt": "2026-06-20T02:12:42.000Z",
  "Stars": [
    {
      "id": 1,
      "name": "Sun",
      "size": 1392000,
      "description": "The star at the center of our solar system.",
      "GalaxyId": 1,
      "createdAt": "2026-06-20T02:13:33.000Z",
      "updatedAt": "2026-06-20T02:13:33.000Z"
    }
  ]
}
```

READ ONE Operation: GET /stars/1

This screenshot demonstrates reading one Star by ID and shows both the related Galaxy and Planet.

```
$ curl -s http://localhost:3000/stars/1 | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
{
  "id": 1,
  "name": "Sun",
  "size": 1392000,
  "description": "The star at the center of our solar system.",
  "GalaxyId": 1,
  "createdAt": "2026-06-20T02:13:33.000Z",
  "updatedAt": "2026-06-20T02:13:33.000Z",
  "Galaxy": {
    "id": 1,
    "name": "Milky Way",
    "size": 100000,
    "description": "The galaxy that contains our solar system.",
    "createdAt": "2026-06-20T02:12:42.000Z",
    "updatedAt": "2026-06-20T02:12:42.000Z"
  },
  "Planets": [
    {
      "id": 1,
      "name": "Earth",
      "size": 12742,
      "description": "Third planet from the Sun.",
      "createdAt": "2026-06-20T02:18:13.000Z",
      "updatedAt": "2026-06-20T02:18:13.000Z",
      "StarsPlanets": {
        "StarId": 1,
        "PlanetId": 1,
        "createdAt": "2026-06-20T02:24:37.000Z",
        "updatedAt": "2026-06-20T02:24:37.000Z"
      }
    }
  ]
}
```

READ ONE Operation: GET /planets/1

This screenshot demonstrates reading one Planet by ID and shows the related Star.

```
$ curl -s http://localhost:3000/planets/1 | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
{
  "id": 1,
  "name": "Earth",
  "size": 12742,
  "description": "Third planet from the Sun.",
  "createdAt": "2026-06-20T02:18:13.000Z",
  "updatedAt": "2026-06-20T02:18:13.000Z",
  "Stars": [
    {
      "id": 1,
      "name": "Sun",
      "size": 1392000,
      "description": "The star at the center of our solar system.",
      "GalaxyId": 1,
      "createdAt": "2026-06-20T02:13:33.000Z",
      "updatedAt": "2026-06-20T02:13:33.000Z",
      "StarsPlanets": {
        "StarId": 1,
        "PlanetId": 1,
        "createdAt": "2026-06-20T02:24:37.000Z",
        "updatedAt": "2026-06-20T02:24:37.000Z"
      }
    }
  ]
}
```

UPDATE Operations: Galaxy, Star, and Planet

This screenshot demonstrates PUT requests updating Galaxy, Star, and Planet descriptions. The verification response confirms the updated Galaxy and nested Star description.

```

→ assignment git:(main) ✕ docker exec -it assign
ment-wdv442-node-1 sh
$ curl -X PUT -H "Content-Type: application/json" --data '{"description
":"Updated galaxy description."}' http://localhost:3000/galaxies/1
{"updated":1}$ curl -X PUT -H "Content-Type: application/json" --data '
{"description":"Updated star description."}' http://localhost:3000/star
s/1
{"updated":1}$ curl -X PUT -H "Content-Type: application/json" --data '
{"description":"Updated planet description."}' http://localhost:3000/pl
anets/1
{"updated":1}$ curl -s http://localhost:3000/galaxies/1 | node -e "let
d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>consol
e.log(JSON.stringify(JSON.parse(d),null,2)))"
{
  "id": 1,
  "name": "Milky Way",
  "size": 100000,
  "description": "Updated galaxy description.",
  "createdAt": "2026-06-20T02:12:42.000Z",
  "updatedAt": "2026-06-20T02:27:06.000Z",
  "Stars": [
    {
      "id": 1,
      "name": "Sun",
      "size": 1392000,
      "description": "Updated star description.",
      "GalaxyId": 1,
      "createdAt": "2026-06-20T02:13:33.000Z",
      "updatedAt": "2026-06-20T02:27:14.000Z"
    }
  ]
}

```

DELETE Operations: Galaxy, Star, and Planet

This screenshot demonstrates DELETE requests for Planet, Star, and Galaxy. Each returns deleted: 1, and the follow-up GET requests return empty arrays.

```

PROBLEMS  TERMINAL  ...
  docker  +  ▾  🗑  ...  |  ✕
→ assignment git:(main) x docker exec -it assignment-wdv442-node
1 sh
$ curl -X DELETE http://localhost:3000/planets/1
{"deleted":1}$ curl -s http://localhost:3000/planets | node -e "let
d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>con
sole.log(JSON.stringify(JSON.parse(d),null,2)))"
[]
$ curl -X DELETE http://localhost:3000/stars/1
{"deleted":1}$ curl -s http://localhost:3000/stars | node -e "let d=
'';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>conso
le.log(JSON.stringify(JSON.parse(d),null,2)))"
[]
$ curl -X DELETE http://localhost:3000/galaxies/1
{"deleted":1}$ curl -s http://localhost:3000/galaxies | node -e "let
d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>co
nsole.log(JSON.stringify(JSON.parse(d),null,2)))"
[]
$ █

```


Association Proof: Star Includes Planet

This screenshot confirms the many-to-many relationship from Star to Planet. The Sun record includes Earth inside its Planets array.

```
→ assignment git:(main) x docker exec -it assignment-wdv442-node-1 sh

$ curl -s http://localhost:3000/stars | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
[
  {
    "id": 1,
    "name": "Sun",
    "size": 1392000,
    "description": "The star at the center of our solar system.",
    "GalaxyId": 1,
    "createdAt": "2026-06-20T02:13:33.000Z",
    "updatedAt": "2026-06-20T02:13:33.000Z",
    "Galaxy": {
      "id": 1,
      "name": "Milky Way",
      "size": 100000,
      "description": "The galaxy that contains our solar system.",
      "createdAt": "2026-06-20T02:12:42.000Z",
      "updatedAt": "2026-06-20T02:12:42.000Z"
    },
    "Planets": [
      {
        "id": 1,
        "name": "Earth",
        "size": 12742,
        "description": "Third planet from the Sun.",
        "createdAt": "2026-06-20T02:18:13.000Z",
        "updatedAt": "2026-06-20T02:18:13.000Z",
        "StarsPlanets": {
          "StarId": 1,
          "PlanetId": 1,
          "createdAt": "2026-06-20T02:24:37.000Z",
          "updatedAt": "2026-06-20T02:24:37.000Z"
        }
      }
    ]
  }
]
```

Association Proof: Planet Includes Star

This screenshot confirms the many-to-many relationship from Planet to Star. The Earth record includes Sun inside its Stars array.

```
$ curl -s http://localhost:3000/planets | node -e "let d='';process.stdin.on('data',c=>d+=c);process.stdin.on('end',()=>console.log(JSON.stringify(JSON.parse(d),null,2)))"
[
  {
    "id": 1,
    "name": "Earth",
    "size": 12742,
    "description": "Third planet from the Sun.",
    "createdAt": "2026-06-20T02:18:13.000Z",
    "updatedAt": "2026-06-20T02:18:13.000Z",
    "Stars": [
      {
        "id": 1,
        "name": "Sun",
        "size": 1392000,
        "description": "The star at the center of our solar system.",
        "GalaxyId": 1,
        "createdAt": "2026-06-20T02:13:33.000Z",
        "updatedAt": "2026-06-20T02:13:33.000Z",
        "StarsPlanets": {
          "StarId": 1,
          "PlanetId": 1,
          "createdAt": "2026-06-20T02:24:37.000Z",
          "updatedAt": "2026-06-20T02:24:37.000Z"
        }
      }
    ]
  }
]
```